

MODULE - II

Quantum Mechanics

Part A Two-Mark Questions

1. Define quantum mechanics and its scope.
2. State Planck's quantum hypothesis.
3. Express the energy of a photon using Planck's relation.
4. Explain the meaning of quantization of energy with an example.
5. Recall the de Broglie relation and illustrate it with a simple example.
6. Compute the de Broglie wavelength of an electron accelerated through a potential difference V .
7. State Heisenberg's uncertainty principle.
8. Express the uncertainty relation between position and momentum mathematically.
9. Interpret the meaning of a wave function ψ .
10. Formulate the normalization condition for a wave function.
11. Write the time-independent Schrödinger wave equation.
12. Write the time-dependent Schrödinger wave equation.
13. Differentiate between eigenvalue and eigenfunction.
14. Derive the expression for energy levels of a particle in a 1D potential box.
15. Justify why the ground-state energy of a particle in a box is not zero.
16. Describe probability density in terms of wave function.
17. Derive the expression for momentum using the de Broglie relation.
18. Identify at least one engineering application of quantum mechanics.
19. Explain the need for normalizing a wave function.
20. Compare classical and quantum mechanics highlighting two limitations of classical mechanics.

Part B – Sixteen-Mark Questions

1. Derive Planck's law of blackbody radiation using quantum hypothesis.
2. Derive and interpret the expression for de Broglie wavelength.
3. Formulate the time-independent Schrödinger wave equation from first principles.
4. Interpret the physical significance of a wave function and justify the normalization condition.
5. Solve the Schrödinger equation for a 1D potential box and obtain the allowed energy eigenvalues.
6. Compare and contrast time-independent and time-dependent Schrödinger equations.

Numerical Problems

1. Calculate the energy of a photon of wavelength 600 nm.
2. Determine the de Broglie wavelength of an electron accelerated through a potential difference of 100 V.
3. Compute the uncertainty in velocity of an electron if its position is measured within 10^{-10} m.
4. Find the first two energy levels of an electron confined in a 1D box of length 1 nm.
5. Calculate the de Broglie wavelength of a proton moving with speed 10^6 m/s.
6. Find the ratio of energies for $n=3$ and $n=1$ states of a particle in a 1D box.
7. Determine the wavelength of an electron accelerated through 500 V.
8. Compute the frequency and wavelength of a photon having energy 4 eV.
9. Apply Heisenberg's principle to calculate the minimum uncertainty in momentum of an electron if uncertainty in position is 1 Å.
10. Calculate the ground state energy of a particle of mass 9.1×10^{-31} kg confined in a box of length 2 nm.